Modeling Synthetic Jets in Crossflow: How Much Geometry Is Enough?

Mechanical & Industrial Engineering
UNIVERSITY OF TORONTO

Howard H. Ho · Ebenezer E. Essel · Pierre E. Sullivan

TURBULENCE RESEARCH LAB

Goal: Finding a lower-cost synthetic-jet model that preserves the key jet-crossflow physics.

The Challenge

- Whole-actuator SJA simulations are accurate but expensive because the diaphragm requires a dynamic mesh.
- Jet-slot-only models are cheaper, but may inject nonphysical momentum during ingestion.

The Core Idea

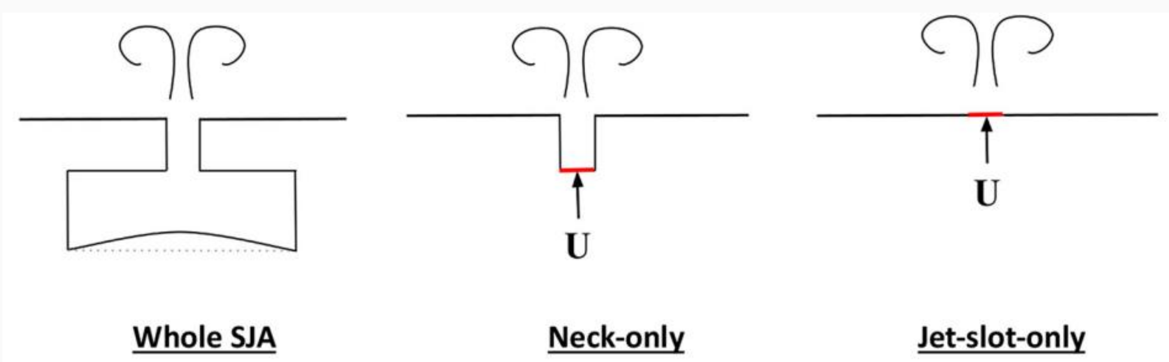
- Compare three circular SJA models in the same crossflow: whole SJA, neck-only, and jet-slot-only.
- Use a Womersley inlet profile for simplified models and compare flow structures, profiles, and cost.

What This Shows

- The neck-only model reproduces the whole-SJA flow field while reducing cost by over 70%.
- A simple volume-ratio rule identifies when a partial model is likely to fail.

1. Modeling choices and numerical setup

SJA Modelling Methods



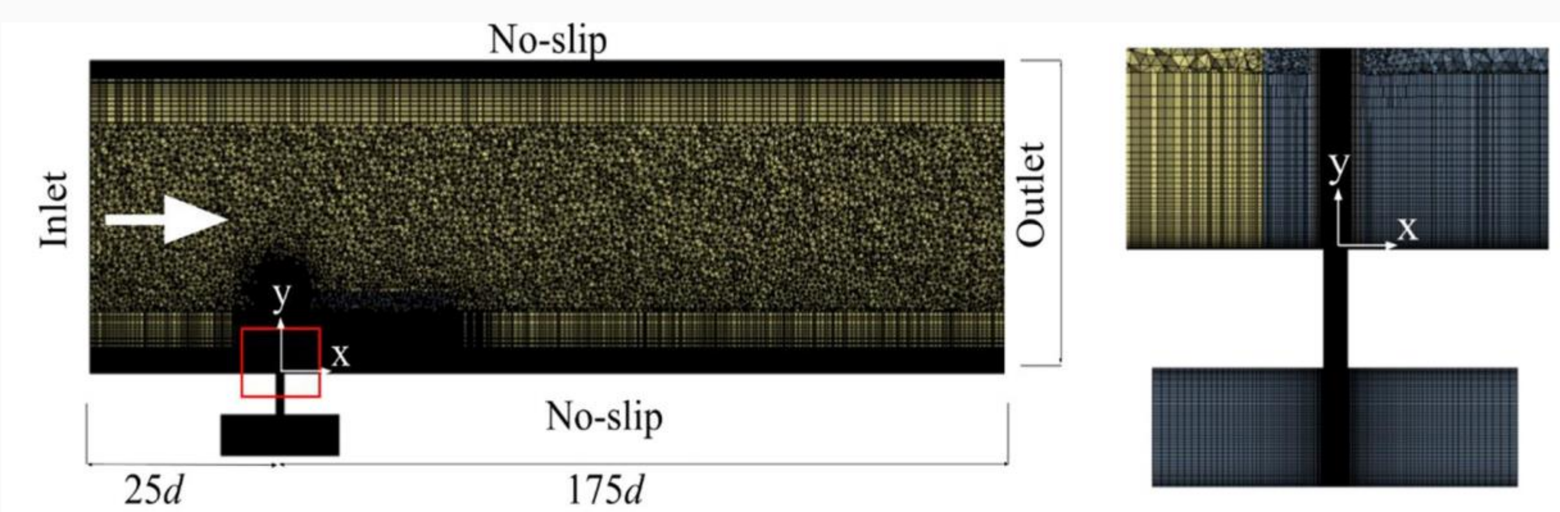
SJA Boundary Conditions

$$s = a\left(1-\left(\frac{r}{r_o}\right)^2\right)\sin(2\pi f t) \quad \text{(Whole SJA)}$$

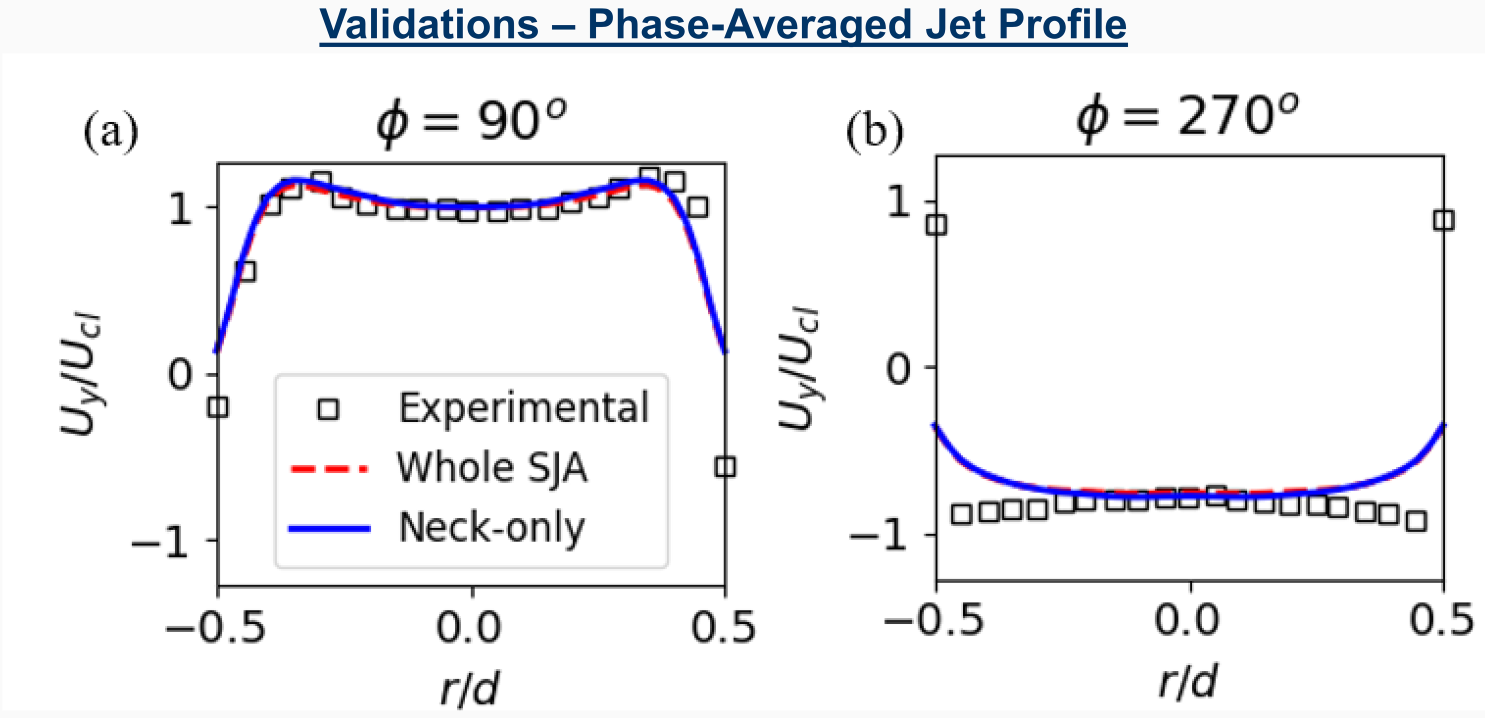
$$v(r,t) = V_j \cdot Real\left\{\left[1-\frac{J_o(i^{3/2}W_o r)}{J_o(i^{3/2}W_o)}\right]e^{i\omega t}\right\} \quad \text{(Neck and Jet-slot Only)}$$

- $Re_\theta = 900$
- SJA diameter: $d = 2$ mm
- $f = 300\ Hz$
- Blowing ratio: $C_B = 0.67$

Computational domain and mesh



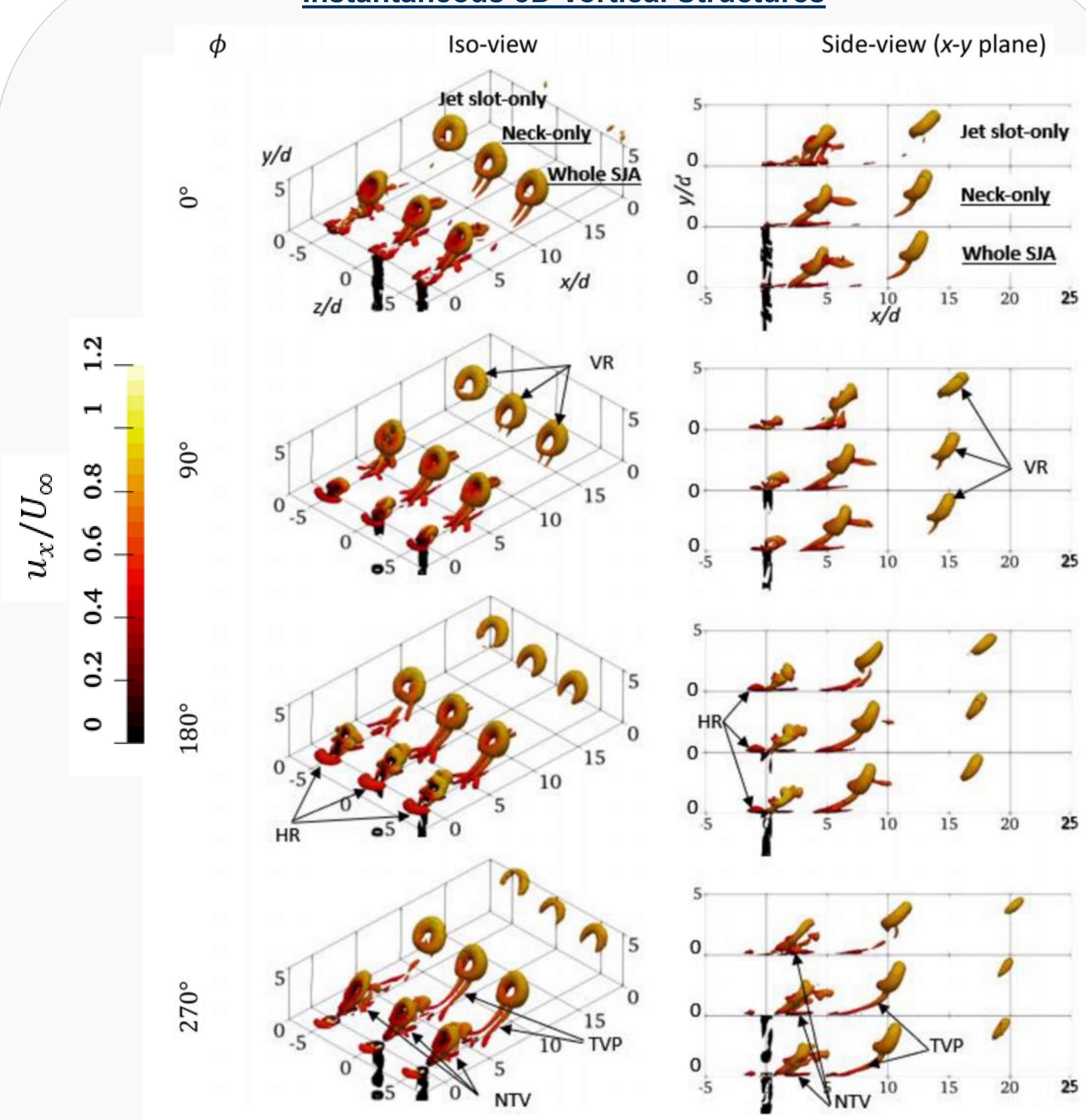
Validations – Phase-Averaged Jet Profile



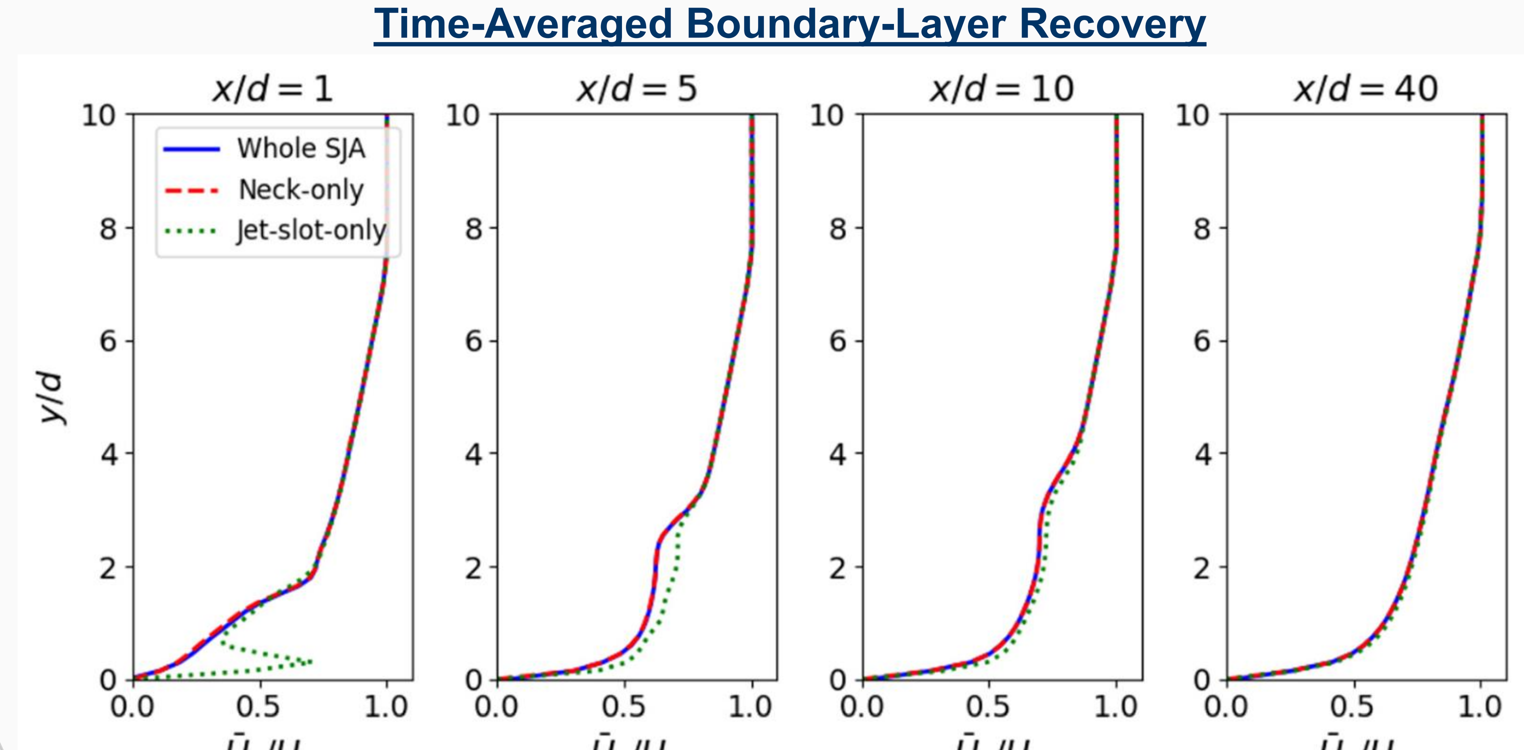
- Quiescent radial jet profiles during peak expulsion and ingestion phase were validated against against hot wire experimental data by Feero et al., 2015.
- The agreement between both models and the experimental data suggests both methods can be used to model SJA, with minimal differences between the whole SJA and neck-only methods in quiescent.

2. Flow-field consequences of simplification

Instantaneous 3D Vortical Structures



Time-Averaged Boundary-Layer Recovery



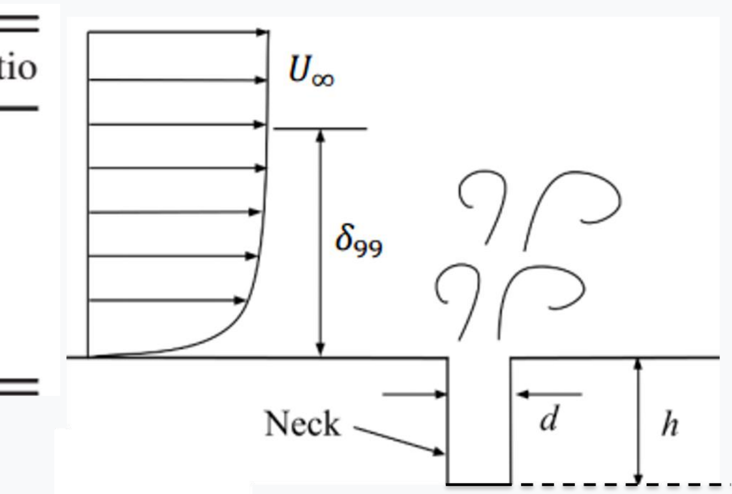
3. Practical modeling rule and recommendation

Results And Discussion

- Whole SJA and neck-only produce nearly identical phase-averaged structures and time-averaged boundary layer profile.
- Jet-slot-only adds a nonphysical expelling jet during ingestion, falsely increasing near-wall momentum and skin friction close to the slot.
- The error decays downstream, but can distort near-wall metrics used to judge separation-control effectiveness.

How Much Neck To Model?

Neck height modeled (h/d)	Volume ratio
0.5	7.58
2	1.89
4	0.95
5 (Whole SJA)	0.76
6	0.63



- The Volume ratio = $\frac{V_{ingested}}{V_{neck}}$
- If Volume ratio $>$ 1 ,the modeled neck is too small to represent ingestion.
- Cases with h/d = 4, 5, and 6 satisfy the rule; h/d = 0.5 and 2 do not.
- The rule was validated with narrow/wide neck cases.

Modeling Guideline

- Use neck-only when the near-neck volume is sufficient: it is cheaper and matches whole-SJA results.
- Avoid jet-slot-only when near-wall shear, skin friction, or ingestion-cycle physics matter.
- For partial SJA models, check the volume ratio before interpreting near-wall momentum or turbulence.

How To Choose A Model

Whole SJA	Best physics reference; dynamic mesh makes it expensive.
Neck-only	Best compromise: matches whole SJA with much lower cost.
Jet-slot-only	Fastest geometry setup, but risky near the jet exit.

Take-Home Message

A neck-only synthetic-jet model can reproduce the whole-actuator flow field at much lower computational cost, provided the modeled neck volume is large enough. Applying the same profile directly at the slot can create false momentum and overpredict near-wall control effects.

References

Feero, M. A., Lavoie, P., and Sullivan, P. E., 2015, "Influence of Cavity Shape on Synthetic Jet Performance," Sensors Actuators, A Phys., **223**, pp. 1–10..

Where This Helps

- Faster parametric studies of SJA placement, frequency, and momentum input.
- More reliable CFD models for active separation control.
- A practical quality check for simplified actuator boundary conditions.

Funding/support: NSERC · CMC Microsystems · SciNet HPC Consortium